INCH-POUND

MIL-DTL-83723/94F
1 October 2010
SUPERSEDING
MIL-DTL-83723/94E
23 May 2003

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, (CIRCULAR, ENVIRONMENT RESISTING), RECEPTACLE, (SINGLE HOLE MOUNT, BAYONET COUPLING, STRAIGHT PIN CONTACT), (SERIES III, CLASSES H, AND Y)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83723.

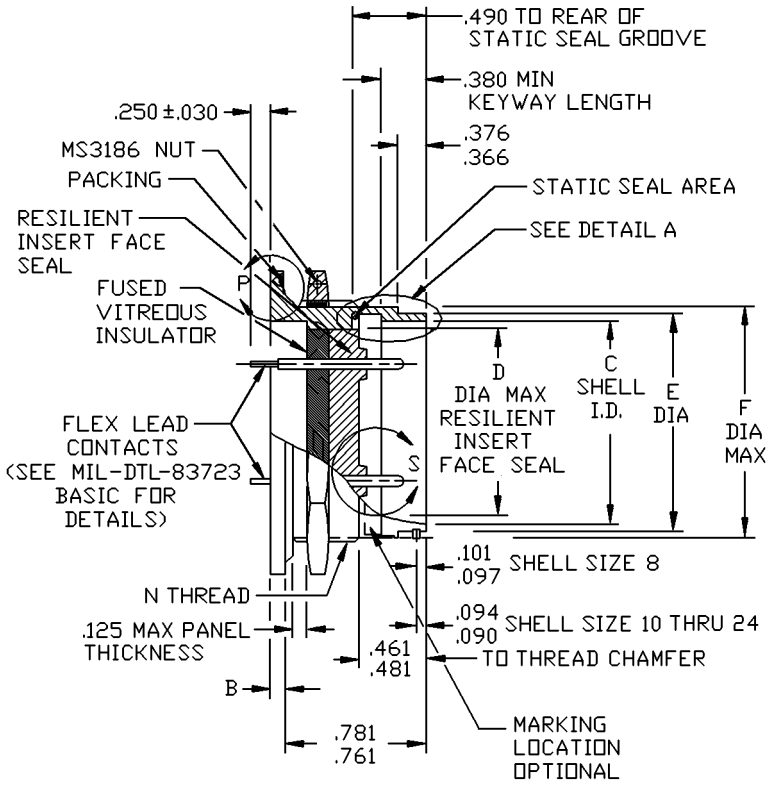


FIGURE 1. Dimensions and configurations.

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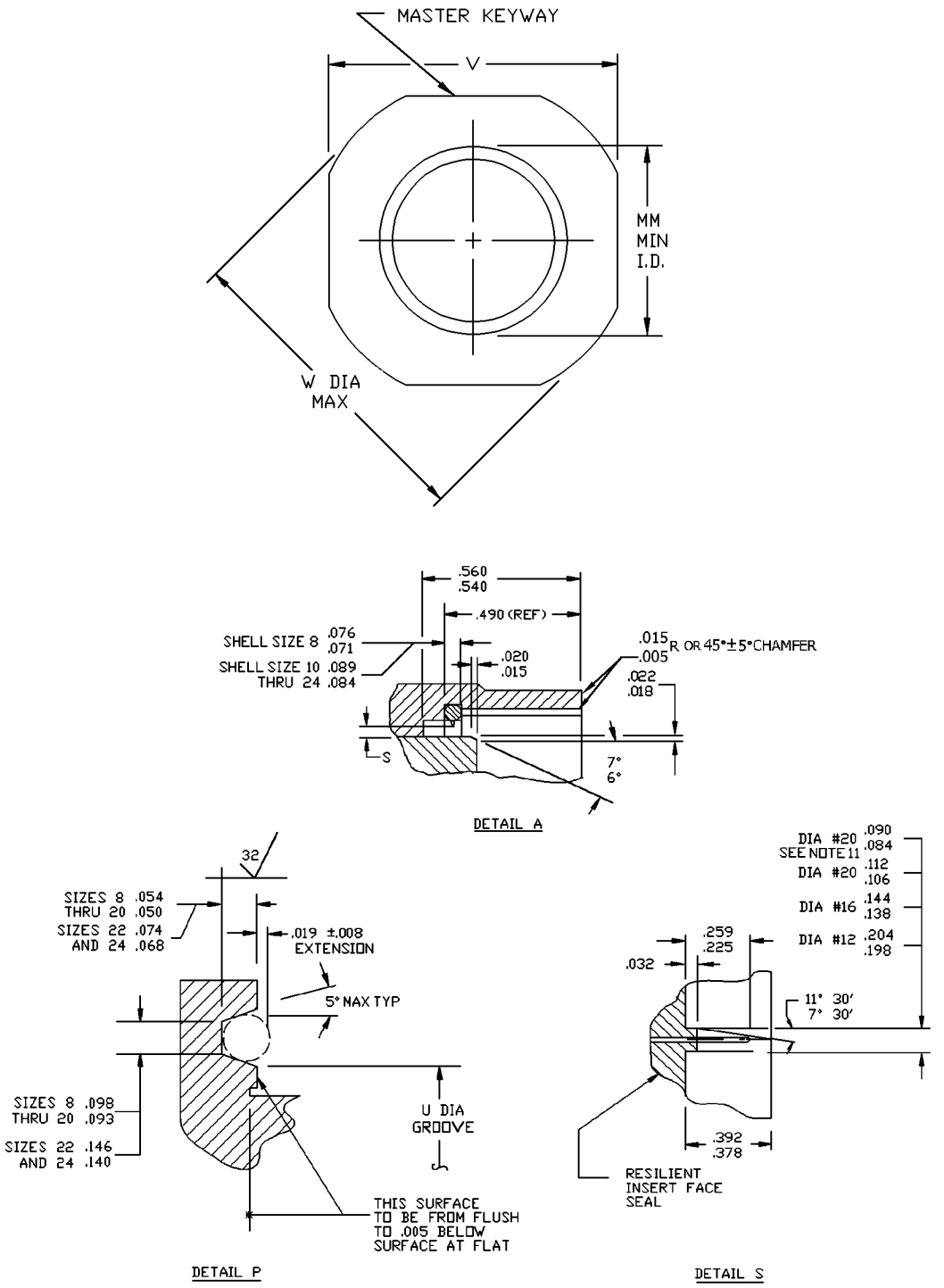
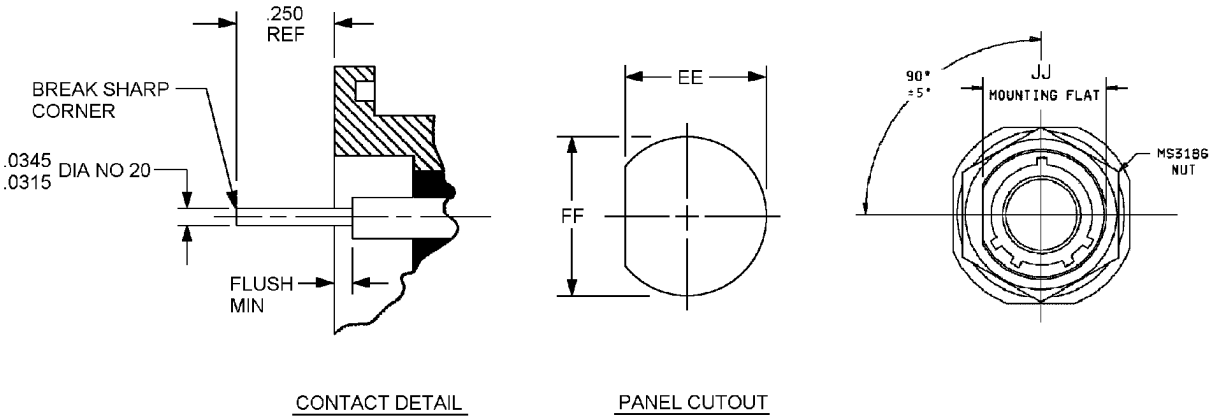


FIGURE 1. Dimensions and configurations – Continued.

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CONTACT DETAIL PANEL CUTOUT

Shell size	B	C	D	E	F	N thread	S +.005 -.000	U
8	.137 .097	.428 .433	.312	.536 .531	.561	.625-20UN-2A	.508	.670
10	.137 .097	.530 .535	.388	.659 .654	.696	.750-20UNEF-2A	.636	.777
12	.137 .097	.700 .705	.558	.829 .824	.875	.9375-20UNEF-2A	.806	.965
14	.137 .097	.769 .774	.627	.898 .893	.935	1.000-20UNEF-2A	.875	1.090
16	.137 .097	.896 .901	.772	1.025 1.020	1.062	1.125-20UNEF-2A	1.002	1.215
18	.137 .097	1.002 1.007	.860	1.131 1.126	1.187	1.250-18UNEF-2A	1.108	1.340
20	.137 .097	1.127 1.132	.985	1.256 1.251	1.312	1.375-18UNEF-2A	1.233	1.465
22	.168 .128	1.252 1.257	1.110	1.381 1.376	1.437	1.500-18UNEF-2A	1.358	1.574
24	.168 .128	1.377 1.382	1.235	1.506 1.501	1.562	1.625-18UNEF-2A	1.483	1.699

FIGURE 1. Dimensions and configurations – Continued.

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Shell size	V	M	EE	FF	MM	JJ
8	.984 .974	1.068	.605	.635	.493	.596 .590
10	1.109 1.099	1.192	.730	.760	.555	.721 .715
12	1.296 1.286	1.380	.917	.947	.743	.908 .902
14	1.396 1.386	1.505	.980	1.010	.805	.971 .965
16	1.521 1.511	1.630	1.105	1.135	.931	1.096 1.090
18	1.646 1.636	1.756	1.229	1.260	1.055	1.220 1.214
20	1.771 1.761	1.860	1.354	1.385	1.243	1.345 1.339
22	1.959 1.949	2.068	1.479	1.510	1.305	1.470 1.464
24	2.084 2.074	2.160	1.604	1.635	1.493 1.452	1.595 1.589

FIGURE 1. Dimensions and configurations – Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.000050	0.0013	.089	2.26	.259	6.58	.558	14.17	1.589	40.36
.003	0.08	.090	2.29	.312	7.92	.560	14.22	1.595	40.51
.004	0.10	.093	2.36	.366	9.30	.561	14.25	1.604	40.74
.005	0.13	.094	2.39	.376	9.55	.590	14.99	1.630	41.40
.008	0.20	.097	2.46	.378	9.60	.596	15.14	1.635	41.53
.010	0.25	.098	2.49	.380	9.65	.605	15.37	1.636	41.55
.015	0.38	.101	2.57	.388	9.68	.627	15.93	1.646	41.81
.018	0.46	.106	2.69	.392	9.96	.635	16.13	1.699	43.15
.019	0.48	.112	2.84	.428	10.87	.636	16.15	1.752	44.50
.020	0.51	.128	3.25	.433	11.00	.654	16.61	1.756	44.60
.022	0.56	.137	3.48	.461	11.71	.659	16.74	1.761	44.98
.030	0.76	.138	3.51	.481	12.22	.670	17.02	1.860	47.24
.032	0.81	.140	3.56	.490	12.45	.696	17.68	1.949	49.50
.050	1.27	.144	3.66	.493	12.52	.700	17.78	1.959	49.76
.054	1.37	.146	3.71	.530	13.46	.705	17.91	2.068	52.53
.068	1.73	.168	4.27	.531	13.49	.715	18.16	2.074	52.68
.071	1.80	.198	5.03	.535	13.59	.721	18.31	2.084	52.93
.074	1.88	.204	5.18	.536	13.61	.730	18.54	2.160	54.86
.076	1.93	.225	5.72	.540	13.72	.743	18.87		
.084	2.13	.250	6.35	.555	14.10	.760	19.30		
.761	19.33	.980	24.89	1.187	30.15	1.376	34.95		
.769	19.53	.984	24.99	1.192	30.28	1.377	34.98		
.772	19.61	.985	25.02	1.214	30.84	1.380	35.05		
.777	19.74	1.002	25.45	1.215	30.86	1.381	35.08		
.781	19.84	1.007	25.54	1.220	30.99	1.382	35.10		
.805	20.45	1.010	25.65	1.229	31.27	1.385	35.18		
.824	20.93	1.020	25.91	1.233	31.31	1.386	35.20		
.829	21.06	1.025	26.04	1.235	31.37	1.396	35.46		
.860	21.84	1.055	26.80	1.243	31.57	1.437	36.20		
.875	22.23	1.062	26.97	1.251	31.78	1.464	37.19		
.893	22.68	1.068	27.13	1.252	31.80	1.465	37.21		
.896	22.76	1.090	27.69	1.256	31.90	1.470	37.47		
.898	22.81	1.096	27.84	1.257	31.93	1.479	37.57		
.901	22.89	1.099	27.91	1.260	32.00	1.483	37.67		
.902	22.91	1.105	28.07	1.286	32.66	1.493	37.92		
.908	23.06	1.108	28.14	1.296	32.92	1.501	38.13		
.917	23.29	1.109	28.17	1.305	33.15	1.505	38.23		
.931	23.65	1.110	28.19	1.312	33.32	1.506	38.25		
.935	23.75	1.126	28.60	1.339	34.01	1.510	38.35		
.947	24.05	1.127	28.63	1.340	34.03	1.511	38.38		
.965	24.51	1.131	28.73	1.345	34.16	1.521	38.63		
.971	24.66	1.132	28.75	1.354	34.39	1.562	39.67		
.974	24.74	1.135	28.83	1.358	34.49	1.574	39.98		

FIGURE 1. Dimensions and configurations – Continued.

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NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ± .005 inches (0.13mm) for three place decimals, ± .01 inches (0.25mm) for two place decimals, and ± 1° for all angles.
4. Notes 5 through 10 applicable to contacts only.
5. Dimensions apply after plating.
6. Contact finish: Gold plate .000050 inch (0.00127 mm) minimum in accordance with ASTM B488. A suitable underplating shall be used. Silver shall not be used. Finish is applied after sealing.
7. Surface finish: 63 ($\sqrt{63}$) all over in accordance with ASME-B46.1 unless otherwise specified.
8. Remove all burrs and sharp edges .003 to .005 inches (0.076 to 0.127 mm) R.
9. Diameters to be concentric within .004 inches (0.102 mm) TIR.
10. Diameters in the same cross-sectional plane to be concentric with each other within .003 (0.076 mm) TIR.
11. See table I for contact cavities requiring this dimension.

FIGURE 1. Dimensions and configurations – Continued.

REQUIREMENTS

Dimensions and configurations: See figure 1.

This connector mates with MIL-DTL-83723/75 and /77.

Insert arrangements and polarizing positions: See MIL-STD-1554 and table I.

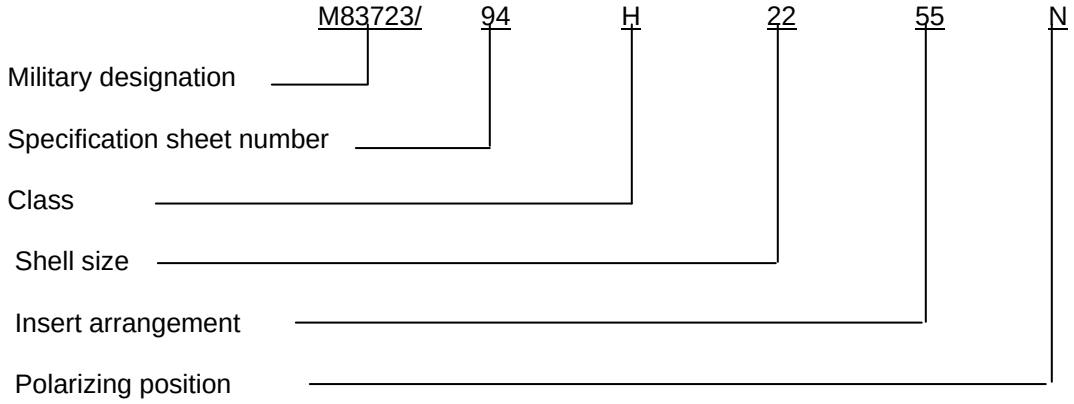
TABLE I. Insert arrangements requiring reduced diameters for raised seal barriers on contact cavities as indicated.

Shell size	Insert arrangements	Contact cavities
8	03, 98	1, 2, 3
10	06	1, 2, 3, 4, 5, 6

Connector accessories: See SAE-AS85049.

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Part or Identifying Number (PIN) example:



Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-83723, this document references the following:

MIL-DTL-83723/75
MIL-DTL-83723/77
MIL-STD-1554
MS3186
ASME-B46.1
ASTM B488
SAE-AS85049

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – AS
Air Force – 85
DLA – CC

Preparing activity:
DLA – CC

(Project 5935–2010-055)

Review activities:
Army – AR, AV, MI
Navy – EC
Air Force – 19, 99

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